6 (a) (i) Fig. 6.1 shows crests of a plane water wave approaching a barrier with a gap.

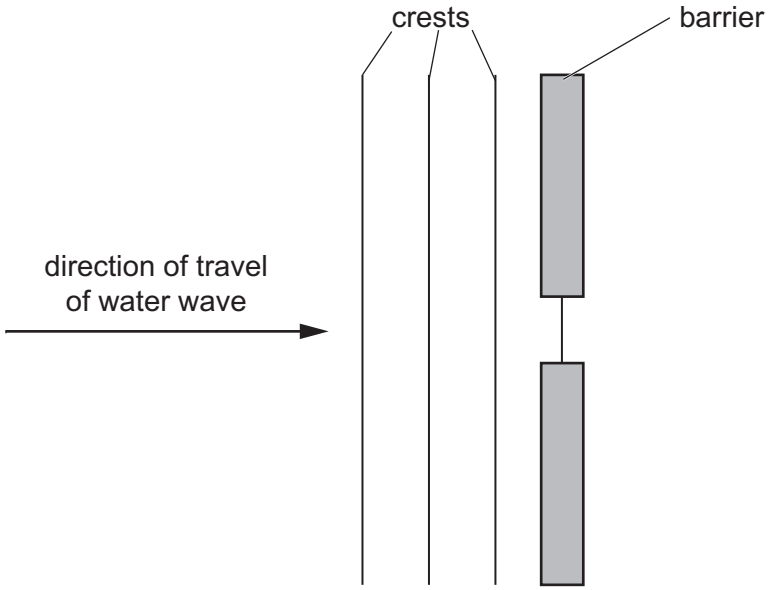


Fig. 6.1

On Fig. 6.1, draw **three** crests of the water wave to the right of the barrier. [2]

(ii) Fig. 6.2 shows crests of a plane water wave in deep water approaching a region of shallow water.

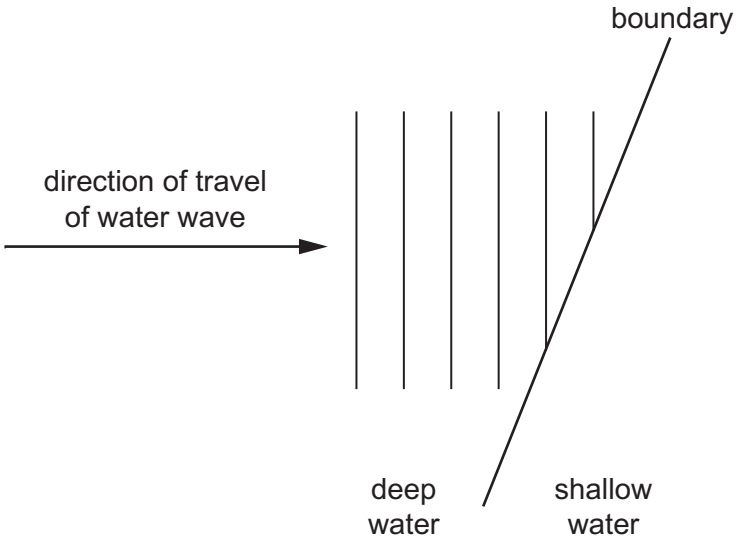


Fig. 6.2

The water wave moves more slowly in shallow water.

On Fig. 6.2, draw:

1. **three** crests of the water wave in the shallow water [2]

2. the direction of travel of the wave in the shallow water. [1]

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(b) State **two** ways in which transverse waves differ from longitudinal waves.

1. .....

.....

2. .....

.....

[2]

(c) **(i)** State a typical value of the speed of sound in water.

..... [1]

(ii) Explain why sound travels faster in water than in air.

..... [1]

[Total: 9]